



OPEN Enhancing wind power forecasting accuracy through LSTM with adaptive wind speed calibration (C-LSTM)

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Wind power constitutes a pivotal component in the quest for carbon neutrality, serving as a principal renewable energy source. Enhancing the accuracy of wind power forecasting facilitates more efficient exploitation of this resource, with deep-learning models, notably Long Short-Term Memory (LSTM), proving effective in advancing forecasting capabilities within this domain. Nevertheless, the accuracy of wind power forecasting is undermined by the inaccurate forecasted wind speed, which diminishes the reliability of such predictions. To address this challenge, we propose the model “LSTM with Adaptive Wind Speed Calibration (C-LSTM)”, which integrates a mechanism into LSTM that autonomously calibrates forecasted wind speed during the training and inference phase. Specifically, considering the inherent continuity of wind speed, C-LSTM fuses historical wind speed with forecasted wind speed using adaptive weighting parameters. This integration is harmonized with the concurrent updating of the other parameters of C-LSTM, thereby ensuring a dynamic adaptation process that bolsters the model’s capacity to coordinate discrepancies between forecasted and actual wind speeds. Experiments conducted across 25 distinct wind turbines have demonstrated that C-LSTM significantly outperforms LSTM in both Mean Squared Error (MSE) and accuracy metrics when the latter directly incorporates forecasted or historical wind speeds. This disparity underscores the efficacy of the adaptive wind speed calibration technique employed within the C-LSTM framework.

Keywords Wind power forecasting, LSTM, Adaptive wind speed calibration, C-LSTM

Abbreviations

LSTM Long Short-Term Memory
C-LSTM LSTM with Adaptive Wind Speed Calibration

Wind power has emerged as a rapidly expanding global energy source, exerting a profound influence on both industrial productivity and daily life¹. Wind energy production is notably distinguished by its intermittent and fluctuating characteristics, which pose significant challenges for energy management and grid stability. The absence of a robust and cost-effective energy storage system mandates the instantaneous utilization of generated power immediately after its production. Consequently, accurate wind power forecasting becomes vital for the proficient management of wind farms and wind-dependent energy systems, facilitating their optimal integration into the energy grid². Accurate forecasting allows for better scheduling of maintenance, more efficient dispatch of conventional power plants, and enhanced overall grid reliability. Nevertheless, the inaccuracy of forecasted wind speed^{3,4}, which stands in contrast to the relative stability of solar radiation in photovoltaic energy systems, amplifies forecasting errors and considerably heightens the complexity of wind power forecasting tasks. These inaccuracies in wind speed forecasts can lead to suboptimal energy dispatch, increased operational costs, and potential disruptions to the balance of supply and demand within the grid. Therefore, developing sophisticated forecasting models that can mitigate these inaccuracies is critical for maximizing the benefits of wind energy and ensuring its seamless integration into the broader energy infrastructure.

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Wind power forecasting encompasses a range of time horizons, including both short-term and long-term predictions⁵. Day-ahead forecasting, which pertains to predictions made for the following day, plays a critical role in the operational planning and market integration of wind farms. This type of forecasting is particularly significant for optimizing the utilization of wind power, as it directly impacts decisions related to energy trading and scheduling within electricity markets⁶. Consequently, this study places a particular emphasis on day-ahead forecasting of wind power. Scholarly interest in wind power forecasting has surged, culminating in the development of a diverse array of forecasting models. These models can largely be categorized into two primary groups: those grounded in domain knowledge (knowledge-based) and those rooted in historical data (data-driven).

Knowledge-based models harness physical principles to simulate and forecast wind power, leveraging deep insights into the fundamental mechanisms governing wind dynamics. Notable examples include innovative smoothing methodologies addressing physical problems associated with wind power⁷, such as turbulence and gustiness, which can significantly impact the performance and reliability of wind turbines. Another example is the use of wavelet decomposition coupled with AdaBoost techniques for forecasting⁸, which decomposes the wind speed signal into different frequency components and applies ensemble learning to improve prediction accuracy. Moreover, sophisticated Numerical Weather Prediction (NWP) frameworks integrating climatic simulations and wind power curve analyses to provide detailed forecasts^{9,10}. These frameworks utilize meteorological data to predict future wind conditions, which are then translated into expected power outputs using turbine-specific wind power curves^{11,12}. These knowledge-based models provide interpretative clarity and a strong theoretical foundation, but they require extensive physical feature extraction and may not fully leverage the richness of historical wind power data.

Conversely, data-driven models leverage historical datasets for forecasting purposes, divided into classical statistical methodologies and deep-learning-based models¹³. Traditional statistical applications, such as autoregressive moving average (ARMA)¹⁴ and autoregressive integrated moving average (ARIMA)¹⁵, excel in short-term predictions due to their ability to model linear dependencies in time series data. However, these models falter when the forecast horizon extends, as they struggle to capture the complex and nonlinear dynamics that are inherent in long-term wind power forecasting. The advent of deep-learning has introduced a suite of advanced models that are capable of capturing intricate nonlinear relationships and have shown significant promise in handling complex time-series data. Examples include the multilayer perceptron (MLP)^{16,17}, which is a feedforward neural network architecture that can learn complex mappings between inputs and outputs. Recurrent Neural Networks (RNNs)^{18–20} are specifically designed to handle sequential data by maintaining an internal state that allows the model to consider past information in its predictions. Variants of RNNs, such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs), have been particularly successful in wind power forecasting due to their ability to manage long-term dependencies effectively. Convolutional Neural Networks (CNNs)^{21,22}, traditionally used in image processing, have also found applications in wind power forecasting, particularly when dealing with spatial-temporal data such as wind maps. CNNs can extract hierarchical features from multidimensional data, making them useful for scenarios where spatial correlations play a significant role. Furthermore, Autoencoders (AEs)^{23–25} are another class of deep-learning models that are adept at unsupervised learning tasks, such as dimensionality reduction and feature learning. They can be used to preprocess the input data, extracting meaningful representations that can then be fed into other predictive models. Some advanced Transformer-based models^{26,27} also show significant potential in wind power forecasting. These pure data-driven deep-learning models are capable of capturing intricate nonlinear relationships and have shown promise in handling complex time-series data. However, they require substantial amounts of data to yield satisfactory results and face the challenge of potentially converging to a local minimum during training, which can hinder the attainment of optimal prediction accuracy. The incorporation of domain knowledge may effectively alleviate these challenges.

Neither knowledge-based models nor data-driven models, when used in isolation, fully address the problem of inaccurate forecasted wind speed. It is recognized that combining these two paradigms can lead to more robust solutions. Indeed, knowledge and data dual-driven models has been widely utilized in multiple domains^{28–32}. In this research, we adopt this hybrid approach to address the issue of inaccuracies in forecasted wind speed and to enhance the accuracy of wind power forecasting. Specifically, based on domain knowledge, it is well-acknowledged that wind speed exhibits a degree of temporal continuity^{33,34}, with the wind speed on any given day typically exhibiting correlation with that of the preceding day, barring instances of extreme weather events. However, due to the complexity inherent in modeling the precise relationship between wind speeds across consecutive days, current forecasted wind speed often fails to adequately incorporate the actual wind speed measurements from the previous day⁴. This oversight in leveraging temporal continuity contributes to a shortfall in the accuracy of forecasted wind speed.

In this study, we introduce C-LSTM, an innovative approach that incorporates an adaptive mechanism within the LSTM³⁵ architecture for calibrating forecasted wind speeds. Unlike traditional methodologies, our model enhances the input to the LSTM by simultaneously incorporating both the actual wind speed from the preceding day and the raw forecasted wind speed for the current day. These two inputs are integrated through learnable parameters, which are updated alongside the other parameters of the C-LSTM network during the training process. This integration allows the model to dynamically adjust its forecasts based on the latest available data, resulting in a calibrated and more accurate forecast of wind speed³⁶. This calibrated forecasted wind speed, along with historical wind speeds, historical wind power, and other pertinent variables, collectively serve as the foundation for the LSTM's modeling process. By leveraging this enhanced input, C-LSTM aims to surmount the challenges posed by the inaccuracy of wind speed forecasts, thereby yielding more precise wind power predictions. The superior performance of C-LSTM has been validated across a diverse set of 25 individual wind turbines, demonstrating its effectiveness in varied conditions and environments. This extensive validation

underscores the model's robustness and generalizability, highlighting its potential for broad application in wind power forecasting. This capability is crucial for grid operators and energy planners who rely on accurate predictions to manage power distribution and ensure grid stability.

In summary, our main contributions of this study include:

- This study introduces a novel deep-learning model C-LSTM, which embeds an adaptive mechanism to calibrate forecasted wind speeds during both training and inference phases. This innovation addresses a crucial limitation in existing wind power forecasting models by enhancing the accuracy of forecasted wind speeds.
- Through extensive experiments on 25 different wind turbines, we validate the superior performance of C-LSTM. The model consistently achieves lower MSE and higher accuracy in comparison to conventional LSTM when confronted with both raw forecasted and historical wind speed data.

Methodology

In this study, we utilize the Long Short-Term Memory (LSTM) model with adaptive wind speed calibration (C-LSTM) to forecast wind power and address the challenge of inaccuracy in raw forecasted wind speeds. This section begins with an exposition of the fundamental deep-learning model, LSTM, which serves as the backbone of the C-LSTM architecture. We then proceed to elucidate the methodology for calibrating forecasted wind speeds and detail how this calibrated data is integrated within the C-LSTM framework.

LSTM

Long short-term memory (LSTM) network³⁵, a classic variant in the realm of neural networks, excels in handling sequential data. It finds applications across various domains, including weather patterns³⁷, audio signals³⁸ and oil well data³⁹.

LSTM is a type of Recurrent Neural Networks (RNNs). RNNs are a class of neural networks designed to recognize patterns in sequences of data, such as time series or natural language. Unlike standard feedforward neural networks such as MLPs, which assume that inputs are independent of each other, RNNs maintain an internal state (hidden state) that allows them to capture information about the past, making them particularly well-suited for tasks involving sequential data. The architecture of an RNN includes loops in the network's connections, which enable information to persist. During the forward pass, the network takes both the current input and the previous hidden state as inputs to compute the current hidden state and the output. This recurrent connection allows the network to have a form of memory, enabling it to use learned information from prior inputs to influence the processing of subsequent inputs.

However, traditional RNNs often struggle to maintain information over long periods due to the issues of vanishing and exploding gradients. The vanishing gradient problem occurs when the gradients of the error function, which are propagated backward through time during training, become very small, leading to minimal changes in the weights of early layers. Conversely, the exploding gradient problem happens when these gradients become excessively large, potentially causing weights to update unpredictably or diverge.

LSTMs address these issues by introducing a more sophisticated internal structure, as illustrated in Fig. 1. Central to this architecture is the cell state C_t , which traverses the LSTM cells, preserving historical information. The LSTM cell also comprises a forget gate f_t , an input gate i_t and an output gate o_t , collaborating alongside the prior hidden state h_{t-1} and the current input X_t to modulate information retention and introduction. Specifically:

The forget gate, represented by Eq. (1), decides what information to discard from the cell state based on the current input and the previous hidden state. The output f_t ranges between 0 and 1, which will be multiplied with the cell state from previous step in Eq. (4).

$$f_t = \sigma [W_f (h_{t-1}, x_t) + b_f]. \quad (1)$$

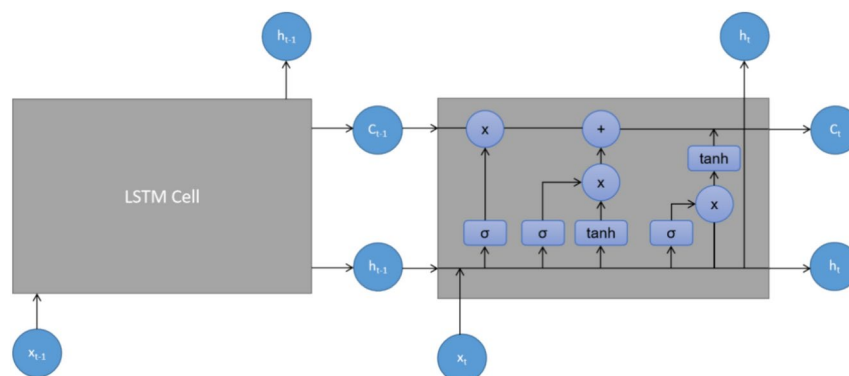


Fig. 1. Structure of LSTM cell. At the core of the LSTM cell is the cell state C_t , which acts as a conduit for carrying historical information through the LSTM cells. The LSTM cell includes a forget gate f_t , an input gate i_t and an output gate o_t . These components work together with the previous hidden state h_{t-1} and the current input X_t to manage the passage and transformation of information.

The input gate is a crucial component responsible for regulating the flow of information into the memory cell. It assesses input data and decides which parts are relevant to retain, as described by Eq. (2). The output i_t is also a number with a range of [0,1], which will be multiplied with the new candidate cell state in Eq. (4).

$$i_t = \sigma [W_i (h_{t-1}, x_t) + b_i]. \quad (2)$$

And Eq. (3) defines the new candidate cell state, incorporating both the hidden state h and the input X :

$$\tilde{C}_t = \tanh [W_c (h_{t-1}, x_t) + b_c]. \quad (3)$$

The final updated cell state C_t results from a combination of the previous cell state and the new candidate cell state, as shown in Eq. (4). Here, the retention of information is controlled by the forget gate f_t and the input gate i_t .

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t. \quad (4)$$

The output gate o_t governs the LSTM cell's output, leveraging the updated cell state C_t to generate the current output h_t , which subsequently serves as both the cell's immediate output and the hidden state transmitted to the succeeding LSTM cell. This comprehensive mechanism is encapsulated in Eq. (5).

$$o_t = \sigma [W_o (h_{t-1}, x_t) + b_o], h_t = o_t \tanh C_t. \quad (5)$$

Despite its complexity, the design of LSTM effectively mitigates gradient issues, enabling it to perform short-term feature extraction similar to that of RNNs while also capturing long-term dependencies²⁹. This dual capability renders LSTM highly suitable for wind power forecasting tasks, where the ability to accurately predict power generation relies on both recent trends and historical patterns. The ability of LSTM to handle such complex sequences makes it an efficient tool for enhancing the precision of wind power predictions, thereby supporting more efficient grid management and renewable energy integration.

Adaptive wind speed calibration in C-LSTM

In the proposed C-LSTM model, we enhance the standard LSTM architecture by integrating an adaptive calibration mechanism specifically tailored for forecasted wind speeds. Recognizing the inherent continuity and temporal correlations present in wind speed data, our approach aims to address the common issue where conventional forecasting models may overlook these characteristics. By incorporating historical wind speed data, the adaptive calibration mechanism refines the raw forecasted wind speeds, ensuring that the predictions are more aligned with real-world patterns. This process is illustrated in Fig. 2, demonstrating how the model leverages historical wind speed data to calibrate the forecasted wind speed.

In the figure, different colors and symbols represent specific aspects of the wind power calibration process: green denotes the raw forecasted wind speed for the current day, gray represents historical wind speed data from the last three days, and blue indicates the historical wind power data from the same period. The symbols t_{pre} , $t-3$, $t-2$, $t-1$, and t correspond to the respective time steps, while w signifies the weight factor applied to the raw forecasted wind speed. The orange arrow indicates the direction of the calibration process, underscoring the significance of historical wind speed data in refining the raw forecasted wind speed to ensure it more accurately reflects real-world conditions, thereby enhancing the overall accuracy of wind power forecasts. The blue arrow represents the direction of the final wind power forecasting, illustrating the effective utilization of the calibrated forecasted wind speed in generating more precise predictions.

The calibration process is mathematically expressed by Eq. (6):

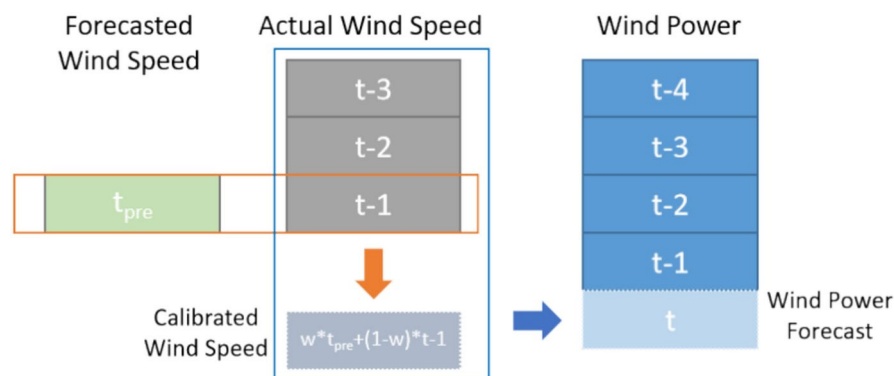


Fig. 2. The forecasted wind speed is calibrated using an adaptive mechanism that incorporates historical wind speed data before being utilized to forecast wind power. This process ensures that the predicted wind speeds are adjusted to more accurately reflect real-world conditions, thereby enhancing the overall accuracy of the wind power forecasts.

$$S_{tc} = WS_{tpred} + (1 - W) S_{t-1} + b, \quad (6)$$

where S_{tc} represents the calibrated forecasted wind speed for the current day, S_{t-1} signifies the actual wind speed from the previous day, and S_{tpred} denotes the raw forecasted wind speed for the current day. W represents the adaptive weighting parameters, and b is the intercept term. Both W and b are 144-dimensional vectors, aligning with the length of a day's wind speed. These parameters are integrated into the LSTM architecture and are iteratively updated during the training phase alongside other model parameters to optimize the overall forecasting accuracy. This adaptive mechanism ensures that the model can dynamically adjust its predictions based on historical data, thereby enhancing the reliability and precision of the wind speed forecasts. During the testing phase, the parameters W and b are fixed at their optimized values obtained during the training phase, ensuring that the calibrated forecast S_{tc} is generated consistently and reliably across new, unseen data.

The inputs to the C-LSTM model are structured in a three-dimensional format: (i) the batch size, which is set at 300 for each training iteration; (ii) the time steps, which utilize 576 steps (corresponding to 4 days of data sampled at 10-min intervals); and (iii) the feature dimensions, which encompass four distinct features—historical wind power, historical wind speed, forecasted wind speed, and turbine pitch angle.

The C-LSTM model comprises two LSTM layers, each configured with a hidden layer size of 20. To augment the model's expressive capacity, two fully connected layers are incorporated, facilitating a richer representation of the input data and contributing to improved forecasting accuracy. This design ensures that the C-LSTM model can effectively capture both temporal dependencies and complex relationships among the input features, thereby enhancing its predictive performance.

Experiment

Data description

In this study, we conducted a case study utilizing wind turbine data from Jiangsu Province, China. The dataset encompasses comprehensive records from the year 2020 for 25 distinct wind turbines, each featuring four key features: pitch angle, historical wind speed, raw forecasted wind speed, and wind power. The relative geographical distribution of these wind turbines is provided in Appendix A. The pitch angle controls the blade angle to optimize energy capture, ensuring that the turbine blades are positioned to harness the maximum amount of wind energy efficiently. Wind speed is a primary driver of turbine operation, directly influencing the rotational speed and, consequently, the power output. Wind power output is the electrical energy generated by the turbines, which is a critical metric for assessing the performance and efficiency of the wind farms. These variables are sampled at a frequency of every 10 min, resulting in approximately 50,000 data points per turbine over the course of the year. This high-resolution dataset provides a detailed view of wind dynamics and power generation patterns.

A visual representation of the feature distributions across these turbines, depicted in Fig. 3, reveals notable similarities among their statistical distributions. These similarities suggest a consistent pattern in the behavior of the turbines, which can be attributed to the shared environmental conditions and operational settings within the same geographical region. This consistency is crucial for developing and validating forecasting models, as it allows for the identification of common trends and the establishment of a reliable baseline for comparative analysis of wind power forecasting.

Data from January to October is used for training, while data from November and December is reserved for testing. The input data spans a duration of four days, encapsulating the features of the pitch angle, historical wind speed, forecasted wind speed, and historical wind power for each turbine. Figure 4 illustrates the comparison between actual and raw forecasted wind speeds over a given period. Specifically, the forecasted wind speeds represent predictions made 24 h in advance. While the overall trend in forecasted wind speeds closely resembles that of the actual wind speeds, discrepancies, particularly in finer details, are notably evident. These discrepancies highlight the challenges in accurately capturing the rapid fluctuations and local variations in wind conditions, which are critical for precise wind power forecasting.

Moreover, Fig. 5 shows wind power curves^{11,40} derived from actual and forecasted wind speeds, highlighting the differences between them. The wind power curve is a graphical representation that depicts the relationship between wind speed and the power output of a wind turbine. Typically, as wind speed increases, so does the power output, up to a point where the turbine reaches its rated capacity. In the figure, the curves derived from actual wind speeds provide a more accurate and smooth representation of the true wind power relationship, reflecting the expected power output under real-world conditions. In contrast, the wind power curves generated from raw forecasted wind speeds exhibit more variability and incorporate a higher degree of noise, which reduces the precision of the power output estimates. This increased noise and variability can be attributed to inaccuracies in the wind speed forecasts, which often fail to fully capture the nuanced and dynamic nature of atmospheric conditions. Such discrepancies highlight the inaccuracies in raw forecasted wind speed and underscore the necessity for calibrating these forecasts.

Experiment setting

In the wind power forecasting experiments, C-LSTM employs a sliding window approach to generate both training and testing samples. This process begins with an initial input window that spans the first four days of historical wind power, pitch angle, historical wind speed, and forecasted wind speed. Since C-LSTM is a day-ahead forecasting model, the forecasting horizon is the future 24 h (144 time steps). Therefore, the corresponding output window includes the wind power data for the subsequent fifth day, forming the first sample. This initial window captures a sufficient temporal context to enable the model to learn the underlying patterns and

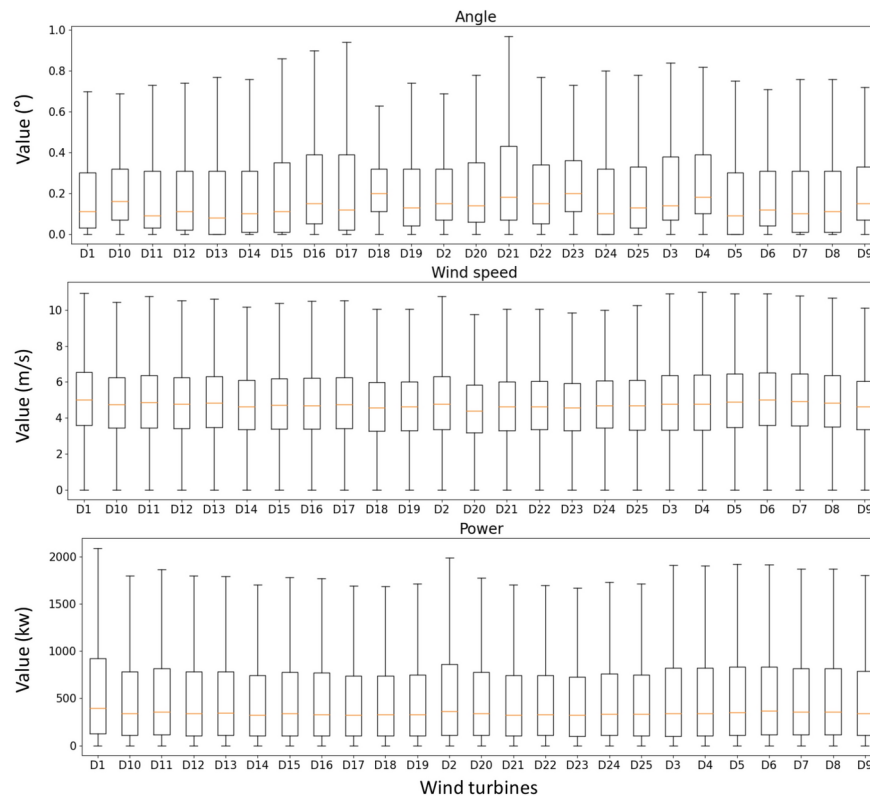


Fig. 3. Pitch angle, wind speed and wind power distribution of wind turbines. The uniformity across different turbines suggests that the pitch angle, wind speed, and wind power attributes exhibit similar characteristics.

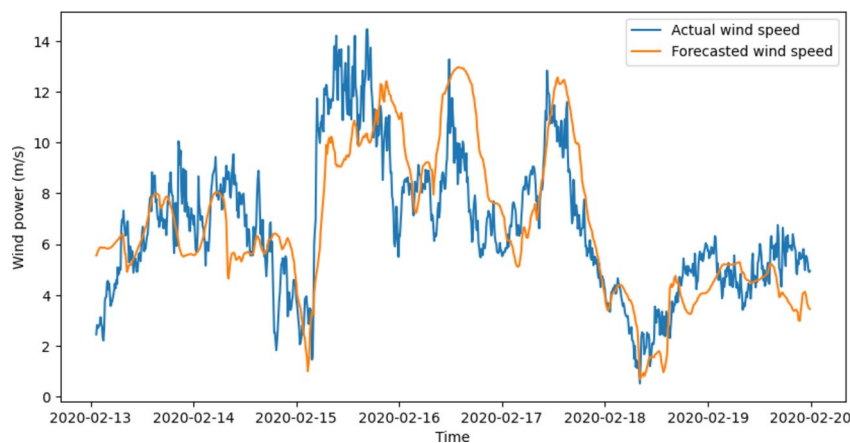


Fig. 4. The comparison of the raw forecasted wind speed and actual wind speed. The forecasted wind speeds represent predictions made 24 h in advance. This figure illustrates the alignment and discrepancies between the forecasted wind speeds and the actual wind speeds over a specified period. While the overall trend in forecasted wind speeds closely resembles that of the actual wind speeds, notable discrepancies, particularly in finer details, are evident.

relationships among the input variables. Additional samples are derived by advancing the input window one day at a time. As the window slides forward, it captures successive days' forecast targets while incorporating new historical context.

Prior to model training, data standardization is initially applied to the input data. Standardization, also known as z-score normalization⁴¹, is a preprocessing technique that transforms the original data distribution into a standardized form, typically with a mean of 0 and a standard deviation of 1. This process ensures that the input features are on a common scale, which is essential for optimizing the performance of machine learning models, particularly those that are sensitive to the scale of the input data, such as LSTMs. By applying standardization,

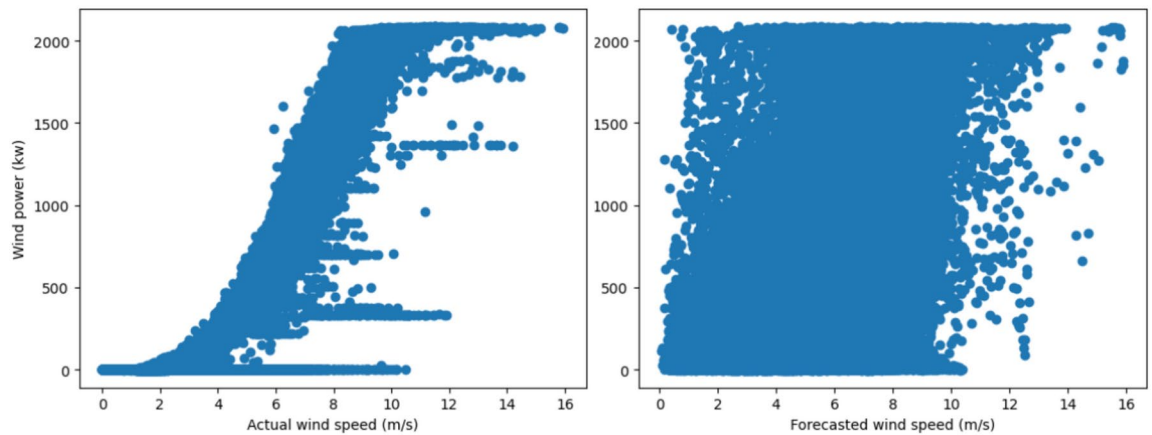


Fig. 5. Wind power curve with actual/forecasted wind speed. The curve based on actual wind speeds provides a more accurate and smoother depiction of the true wind power relationship, while the curve generated from forecasted wind speeds exhibits more variability and incorporates a higher degree of noise.

the model can better learn the underlying patterns in the data without being biased by the scale of the input variables. This is particularly beneficial in the context of wind power forecasting, where variables such as wind speed and wind power can vary widely in magnitude. Standardizing these variables ensures that the model weights are not dominated by the scale of any single variable, leading to improved convergence and stability during training. In practice, the standardization parameters (mean and standard deviation) are calculated using the training dataset and then applied consistently to both the training and testing datasets to ensure that the model is evaluated under similar conditions as it was trained.

For evaluation, mean square error (MSE), accuracy metrics, normalized root mean square error (N_RMSE) and normalized mean absolute error (N_MAE) are used to evaluate the models. The MSE is a quantitative measure of the average squared difference between the forecasted and observed wind power values, defined as follows:

$$MSE = \frac{1}{N} \sum_{i=1}^n (G_i - P_i)^2, \quad (7)$$

where G_i is the actual standardized wind power at time step i , and P_i refers to the standardized predicted wind power. The accuracy metric is used to assess relative discrepancy between forecasted and observed wind power, formulated as follows:

$$Acc = \frac{1}{N} \sum_{i=1}^n \left(1 - \frac{|G_i - P_i|}{Max(G)} \right), \quad (8)$$

where G_i denotes the actual wind power at time step i rescaled to its original scale, P_i represents the predicted wind power rescaled to its original scale, $Max(G)$ is the maximum wind power of the turbine, and N is the number of samples. The N_RMSE metric is defined as:

$$N_RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^n (G_i/C - P_i/C)^2}, \quad (9)$$

where G_i is the actual wind power at time step i , P_i denotes the predicted wind power rescaled to its original scale, C is the installed capacity of the turbine, and N is the number of samples. Using these same symbols, the N_MAE metric can be expressed as:

$$N_MAE = \frac{1}{N} \sum_{i=1}^n |G_i/C - P_i/C|. \quad (10)$$

These metrics together offer a comprehensive evaluation of the model's predictive performance, with the MSE, N_RMSE and N_MAE capturing the magnitude of the errors and the accuracy metric providing insight into the relative discrepancy of the predictions to the actual values.

The training batch size is set to 300, and the training epoch is set to 50, with an early stopping patience of 5 to prevent overfitting. The widely-used MSE loss function is employed, and optimization is performed using the Adam optimizer, which is known for its efficiency and performance in handling sparse gradients and noisy data. Each experimental result is averaged from five independent runs to ensure robustness and reliability of the outcomes. The experiments are conducted on a single NVIDIA GeForce RTX 4090 GPU with a memory capacity of 24 GB, providing the computational capacity necessary for efficient model training and evaluation.

Wind power forecasting results

The test set results of C-LSTM are presented in Table 1. For comparison, several baseline configurations within the LSTM framework were evaluated to assess the effectiveness of the C-LSTM approach. Specifically, these configurations included using forecasted wind speeds directly (labeled as ‘Forecasted’ in the table), incorporating historical wind speeds without modification (referred to as ‘Historical’), and implementing a balanced linear combination of forecasted and historical wind speeds (with each contributing 50%, labeled as ‘Combined’) as inputs to the LSTM. The table lists the average accuracy, MSE, N_RMSE and N_MAE across all 25 wind turbines for each of these configurations. Additionally, the table provides the best and worst accuracy and MSE achieved among the turbines, offering a comprehensive overview of the performance across the entire dataset. The inclusion of the best and worst accuracy, MSE, N_RMSE and N_MAE scores further aids in understanding the variability in performance across the different wind turbines. This information is crucial for identifying any outliers or specific conditions under which the models perform exceptionally well or poorly, which could inform future improvements and refinements in the C-LSTM model.

Table 1 demonstrates that C-LSTM outperforms the other baseline configurations in both accuracy and MSE metrics. Specifically, the incorporation of an adaptive wind speed calibration mechanism within the C-LSTM framework improves accuracy by 0.4% and reduces the MSE, N_RMSE and N_MAE by 0.012, 0.013 and 0.010 compared to using uncalibrated forecasted wind speeds. This enhancement indicates that the dynamic adjustment of forecasted wind speeds based on real-time data contributes significantly to the precision of wind power predictions. When compared to the use of historical wind speed data alone, C-LSTM further enhances accuracy by 0.5% and reduces the MSE, N_RMSE and N_MAE by 0.019, 0.016 and 0.012. This improvement underscores the value of combining historical data with adaptive, calibrated forecasts to achieve a more robust predictive model. Historical data provides a stable reference point, but it lacks the dynamic adjustment capabilities that C-LSTM introduces. Notably, a simplistic linear combination of forecasted and historical wind speeds, with each contributing equally (the ‘Combined’ approach), does not effectively calibrate forecasted winds, as evidenced by its inferior performance relative to uncalibrated forecasted wind speeds. This finding highlights the limitations of fixed-weight averaging strategies and underscores the superiority of C-LSTM’s adaptive calibration mechanism. The learnable parameters within C-LSTM allow for a more nuanced and context-aware adjustment of forecasted speeds, leading to better overall performance. C-LSTM’s improved accuracy and reduced error rate are essential for optimizing wind farm operations, enhancing grid stability, and ensuring that renewable energy sources are integrated effectively into the energy system.

Furthermore, the boxplots of accuracy for the different methods across 25 wind turbines are shown in Fig. 6. The visual representations clearly demonstrate that C-LSTM not only achieves more accurate predictions but also exhibits greater stability in its forecasts. The narrower spread of the C-LSTM boxplot indicates a more consistent performance, with fewer outliers and a tighter clustering around the median, suggesting that the model’s predictions are consistently reliable. In contrast, the wider spread and presence of outliers in the boxplots of the baselines indicate less consistent performance, highlighting the superiority of C-LSTM in both accuracy and stability.

For comparative purposes, we also examined scenarios using actual wind speed data (which is unattainable in real-time deployment) and scenarios where wind speed data is not utilized, relying solely on historical wind power to perform autoregressive forecasting, as depicted in Table 2. The table reveals that utilizing ideal, real-time wind speed information enables remarkably precise wind power forecasts, achieving the highest accuracy and lowest MSE, N_RMSE and N_MAE among all tested configurations. This ideal scenario, while impractical for operational use, sets a benchmark for the maximum achievable forecasting accuracy. Conversely, relying solely on historical wind power for autoregressive forecasting leads to a significant decline in the model’s predictive performance, with higher MSE, N_RMSE and N_MAE values indicating greater discrepancies between predicted and actual wind power. These findings further emphasize the critical role of accurate wind speed in forecasting wind power and underscore the necessity of precise wind speed calibration to enhance forecasting performance.

	MSE	ACC	N_RMSE	N_MAE	MSE	ACC	N_RMSE	N_MAE
	C-LSTM				Forecasted			
Avg	0.437	0.913	0.179	0.127	0.449	0.909	0.192	0.137
Best	0.381	0.923	0.156	0.106	0.387	0.922	0.166	0.116
Worst	0.488	0.898	0.207	0.145	0.685	0.892	0.239	0.167
	Historical				Combined			
Avg	0.456	0.908	0.195	0.139	0.445	0.909	0.191	0.136
Best	0.397	0.922	0.174	0.116	0.404	0.922	0.177	0.120
Worst	0.630	0.896	0.225	0.158	0.602	0.897	0.219	0.152

Table 1. Wind power forecasting experimental results of C-LSTM and baselines. The column labeled “C-LSTM” denotes our proposed model with adaptive wind speed calibration, “Forecast” indicates the use of raw forecasted wind speeds, “Historical” refers to the use of historical wind speeds without modification, and “Combined” represents a balanced linear combination of forecasted and historical wind speeds. The evaluation uses four metrics: accuracy, MSE, N_RMSE, and N_MAE. The unit of accuracy is %, while the other metrics are calculated on standardized or normalized data and are thus dimensionless. Bold font represents the superior result.

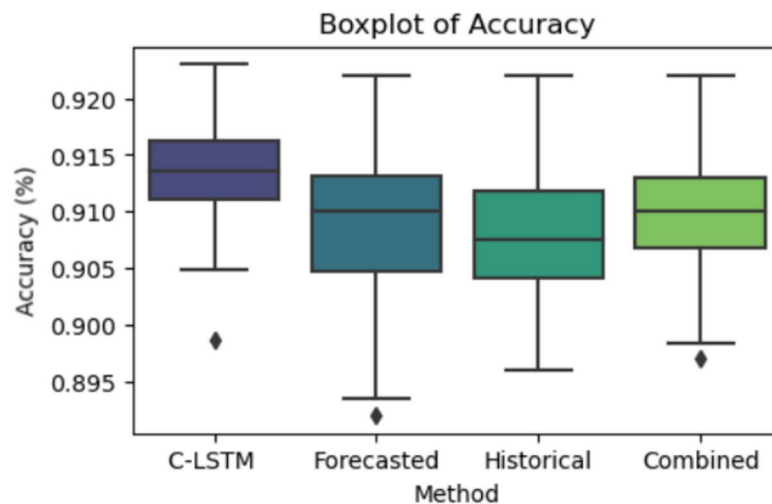


Fig. 6. Comparison of accuracy for different methods. The boxplot illustrates the distribution of accuracy across 25 wind turbines, demonstrating the consistency and reliability of C-LSTM compared to other methods.

	Actual wind speed				No wind speed			
	MSE	ACC	N_RMSE	N_MAE	MSE	ACC	N_RMSE	N_MAE
Avg	0.057	0.969	0.049	0.033	0.588	0.897	0.202	0.143
Best	0.037	0.974	0.041	0.028	0.497	0.912	0.186	0.129
Worst	0.092	0.959	0.064	0.043	0.836	0.880	0.245	0.169

Table 2. Wind power forecasting experimental results using actual wind speed data and autoregressive forecasting without wind speed data. The evaluation uses four metrics: accuracy, MSE, N_RMSE, and N_MAE. The unit of accuracy is %, while the other metrics are calculated on standardized or normalized data and are therefore dimensionless. The results underscore the importance of accurate wind speed data in achieving reliable forecasting performance.

Figure 7 showcases the predictive performance of C-LSTM on a specific wind turbine over a certain period, including magnified views of several intervals. For comparison, the figure also displays the results of the baseline LSTM using raw forecasted wind speeds. The blue line represents the actual wind power values, the green line denotes the predicted values generated by the C-LSTM model, and the red line represents the predictions from the baseline LSTM. Evidently, the forecasts produced by C-LSTM more closely mirror the actual wind power trends, exhibiting exceptional alignment and capturing subtle deviations in wind power fluctuations. This visual representation underscores the model's capability to provide predictions that are not only directionally accurate but also detailed in reflecting the dynamic changes inherent in wind power, thereby validating the effectiveness of the adaptive wind speed calibration mechanism employed by C-LSTM. The close match between the actual and predicted values demonstrates the model's robustness and reliability in forecasting, which is essential for effective wind farm management and grid integration.

Additionally, an intersection chart based on the ground truth and predictions of C-LSTM is shown in Fig. 8. The proximity of the data points in the intersection graph to the 45-degree diagonal indicates the accuracy of C-LSTM; the closer they are, the higher the accuracy. As can be observed from the figure, the intersection of the true values and the predictions of C-LSTM closely aligns with the 45-degree diagonal, indicating that C-LSTM achieves high prediction accuracy.

Furthermore, we have plotted the wind power curve derived from the calibrated wind speed and wind power data, as shown in Fig. 9. Although the calibrated wind power curve is not as smooth as the actual wind power curve (as seen in the left subplot of Fig. 5), its performance is noticeably better than the wind power curve obtained using raw forecasted wind speed (as shown in the right subplot of Fig. 5). This comparison highlights the effectiveness of the adaptive calibration mechanism in enhancing the quality of the forecasted wind speed.

The adaptive wind speed calibration is a key differentiator from the standard LSTM, which directly uses raw forecasted wind speeds without adjustments, and this calibration mechanism is crucial for achieving better wind power forecasting performance. The calibration mechanism is designed to account for the temporal continuity and correlations in wind speed data, enabling the model to automatically adjust the influence of the forecasted wind speed based on historical wind speed. This reduces the impact of inaccuracies in the raw forecasted wind speeds. The effectiveness of this wind speed calibration is demonstrated by the comparison between Fig. 9 and the right subplot of Fig. 5, where the more accurate calibrated wind speeds lead to more precise power predictions. In contrast, the standard LSTM lacks this adaptive calibration mechanism and thus relies solely on the raw

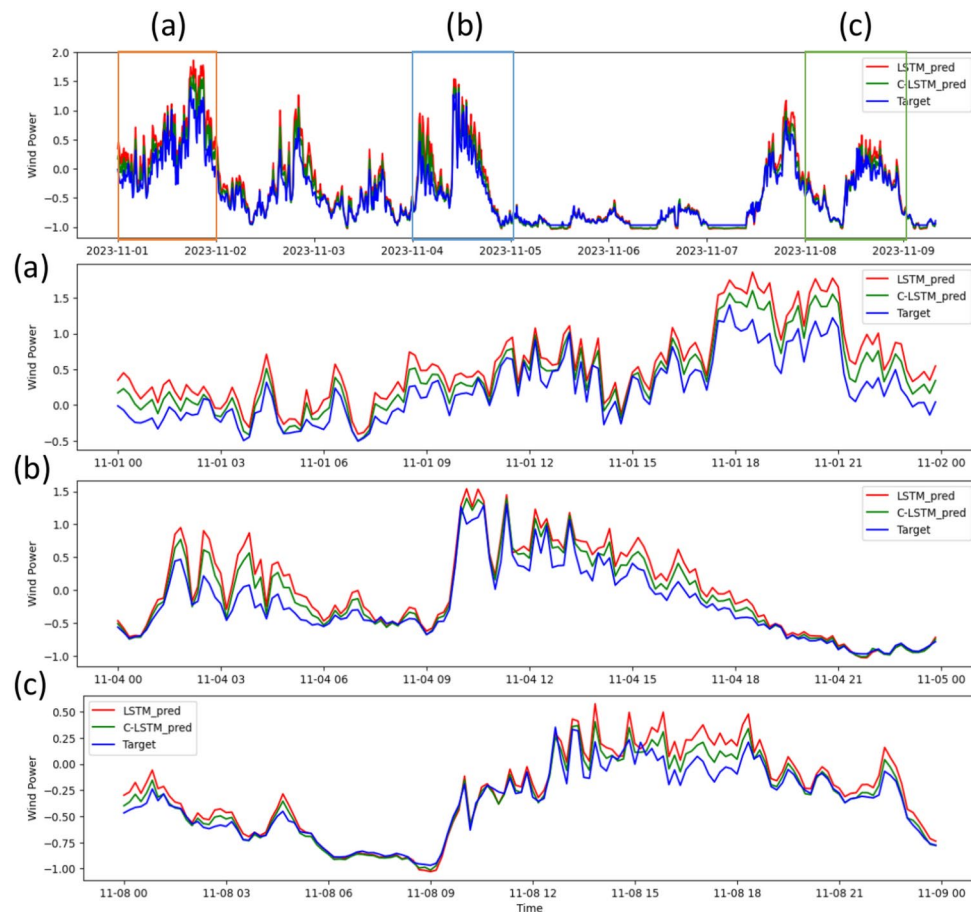


Fig. 7. Showcases of C-LSTM. This visual representation underscores C-LSTM's capability to provide predictions that are not only accurate in trends but also detailed in reflecting the dynamic changes inherent in wind power, compared to the baseline LSTM model.

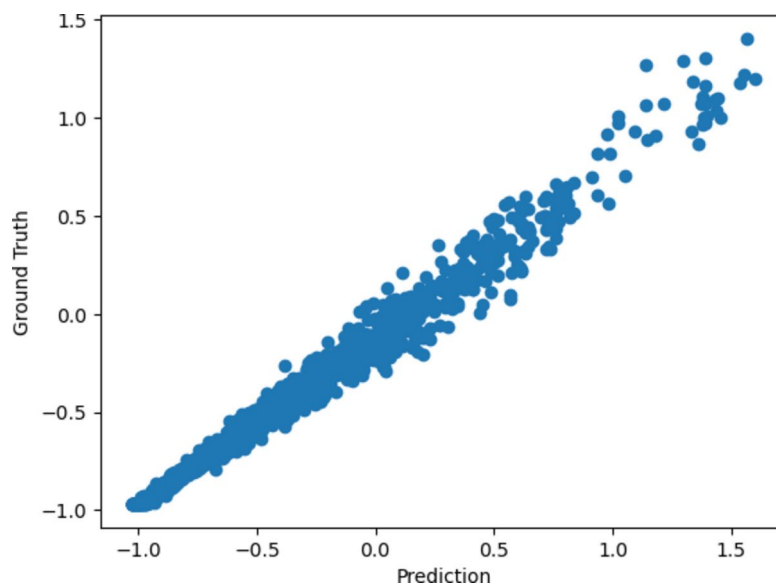


Fig. 8. Intersection chart based on the ground truth and predictions of C-LSTM. The proximity of the data points in the intersection graph to the 45-degree diagonal indicates C-LSTM's accuracy; the closer they are, the higher the accuracy.

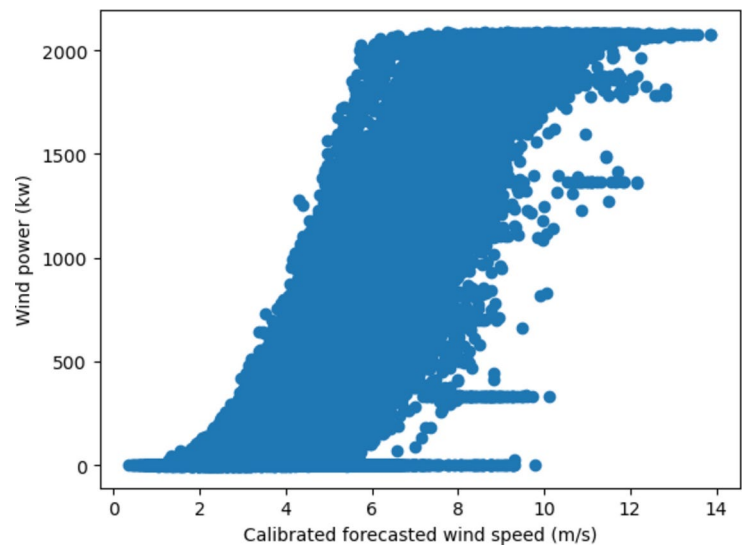


Fig. 9. Wind power curve with forecasted calibrated wind speed. The wind power curve generated from the calibrated forecasted wind speed is smoother compared to the wind power curve generated from the raw forecasted wind speed.

	MSE	ACC	N_RMSE	N_MAE	MSE	ACC	N_RMSE	N_MAE
	C-LSTM				Forecasted			
Avg	0.437	0.913	0.179	0.127	0.449	0.909	0.192	0.137
Best	0.381	0.923	0.156	0.106	0.387	0.922	0.166	0.116
Worst	0.488	0.898	0.207	0.145	0.685	0.892	0.239	0.167
	Weekly				Monthly			
Avg	0.451	0.910	0.193	0.137	0.449	0.909	0.191	0.137
Best	0.389	0.921	0.168	0.117	0.388	0.922	0.166	0.117
Worst	0.682	0.893	0.237	0.166	0.686	0.891	0.241	0.169

Table 3. Adaptive wind speed calibration with the wind speed data from a week or a month ago. The column labeled “C-LSTM” denotes our proposed model with adaptive wind speed calibration with the actual wind speed from the previous day, “Forecast” indicates the use of raw forecasted wind speeds, “Weekly” refers to the wind speed calibration with the actual wind speed from one week ago, and “Monthly” represents the wind speed calibration with the actual wind speed from one month ago. The evaluation is conducted using four metrics: accuracy, MSE, N_RMSE, and N_MAE. The unit of accuracy is %, while the other metrics are calculated on standardized or normalized data and are therefore dimensionless. Bold font represents the superior result.

forecasted wind speeds, which can be erroneous and result in suboptimal predictions. We also apply this wind speed calibration mechanism to other wind power forecasting models to further demonstrate its effectiveness in the following section.

Additional experiments of wind power forecasting

In this section, to further demonstrate the effectiveness of C-LSTM, we have conducted additional experiments, which include an exploration of weekly and monthly correlations, as well as the application of our wind speed calibration method to other effective wind speed forecasting models. Specifically, C-LSTM initially utilized only the actual wind speed from the previous day for calibration. Considering the potential periodicity in wind speed, we also attempted to use wind speed data from one week and one month prior for calibration. However, the results presented in Table 3 indicate that the wind speed data from one week or one month ago did not help the calibration of the forecasted wind speed, and the wind power forecasting performance remained similar to that before calibration. We hypothesize that this is primarily because the wind speed data from one week or one month ago is too temporally distant to provide meaningful improvements in wind speed calibration and wind power forecasting.

To demonstrate the generalizability of our adaptive wind speed calibration method, we also applied the method to other efficient wind power forecasting models, including Bi-LSTM⁴², CNN-LSTM⁴³, Transformer⁴⁴, and a physical model, NWP¹⁰. Table 4 presents the average accuracy, MSE, N_RMSE, and N_MAE of these

	Wind speed calibration				No wind speed calibration			
	MSE	ACC	N_RMSE	N_MAE	MSE	ACC	N_RMSE	N_MAE
Bi-LSTM	0.436	0.913	0.180	0.127	0.451	0.909	0.192	0.138
CNN-LSTM	0.446	0.911	0.182	0.130	0.455	0.908	0.196	0.140
Transformer	0.470	0.905	0.188	0.136	0.491	0.897	0.201	0.147
NWP	0.693	0.881	0.223	0.176	0.695	0.880	0.231	0.180

Table 4. Adaptive wind speed calibration with other four wind power forecasting model: Bi-LSTM, CNN-LSTM, Transformer, and NWP. The evaluation uses four metrics: accuracy, MSE, N_RMSE, and N_MAE. The accuracy’s unit is %, while the other metrics are calculated on standardized or normalized data and are thus dimensionless. Bold font represents the superior result.

models across 25 turbines, both with and without adaptive wind speed calibration. The test results show that our adaptive wind speed calibration is also effective for these models, which further strengthens its reliability.

Conclusion

This study introduces C-LSTM, a pioneering model that integrates an adaptive calibration mechanism within the traditional LSTM architecture to calibrate forecasted wind speeds. By dynamically fusing historical and forecasted wind speeds through trainable parameters, C-LSTM significantly enhances wind power forecasting accuracy. Experiments conducted across 25 diverse wind turbines confirm C-LSTM’s outstanding performance, boosting accuracy by 0.4% and reducing MSE, N_RMSE and N_MAE by 0.012, 0.013 and 0.010 compared to uncalibrated forecasts. Compared to the use of historical wind speed data alone, it further increases accuracy by 0.5% and cuts MSE, N_RMSE and N_MAE by 0.019, 0.016 and 0.012. These improvements highlight the effectiveness of C-LSTM’s adaptive approach in refining wind speed predictions. The model’s success in wind power forecasting not only underscores its potential for enhancing grid management and operational efficiency in wind farms but also suggests broader applicability. C-LSTM’s approach to calibrating weather forecasts holds potential for transferability to other applications, such as photovoltaic power and electricity demand forecasting, thereby contributing to more comprehensive and reliable renewable energy integration strategies.

Data availability

The datasets used and/or analyzed during the current study available from the corresponding author on reasonable request.

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Author contributions

Ding Wang and Min Xu wrote the main manuscript text and Zhu Guangming carried out the processing of the data. Futao Luo discussed the methodology. Jiaxia Gao prepared Figs.1–3 Chen made a valid suggestion. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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